RCE-2 v6.15

Online Causal Refinement — Real-Time Streaming Estimation

v6.15 replaces batch causal estimation with online streaming updates. As new observations arrive, causal estimates update incrementally in O(d) per observation — no full batch re-estimation needed. The substrate now maintains an up-to-date causal model in production, with drift detection via EWMA and convergence guarantees.

Project	Omega-X v6.15
Document	RCE-2 v6.15 Online Causal Refinement Report
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Status	[MEASURED] [IMPLEMENTED] [TESTED]
Test Suite	215 tests (197 prior + 18 online), all passing
Complexity	O(d) per observation (vs O(n*d) for batch)
Key Result	Real-time causal model adaptation with drift detection

1. Executive Summary

This report documents the v6.15 Online Causal Refinement layer. The v6.14 observational causal estimator runs in BATCH: collect all observations, run PSM/IPW/DR, get estimates. When new observations arrive, the entire batch must be re-run — O(n*d) per update, too expensive for production streaming data.

The v6.15 OnlineCausalRefiner updates causal estimates INCREMENTALLY as new observations arrive. It uses streaming logistic regression for propensity scores (O(d) per update), incremental running sums for DR estimation (O(1) per update), and EWMA-based drift detection. The substrate now maintains an up-to-date causal model in real-time, adapting to distribution shifts without expensive batch re-estimation.

Key Results

Metric	Value	Notes
O(d) per observation update	Verified	vs O(n*d) for batch
Streaming logistic regression	Verified	SGD updates propensity scores
Incremental DR estimation	Verified	Running sums, O(1) per update
SE decreases with more data	Verified	0.132 -> 0.096 -> 0.066
ACE converges to true value	Verified	500 obs: |ACE - 1.0| < 0.2
Drift detection via EWMA	Verified	Flags significant estimate changes
1000 observations handled	Verified	Efficient streaming
Test suite	215 tests	18 new online tests, all passing

2. Streaming Causal Estimation

The OnlineCausalRefiner maintains causal estimates incrementally. Three key techniques enable O(d) per-observation updates:

2.1 Streaming Logistic Regression

```
For each new (x, t):
  pred = sigmoid(w . x)
  error = pred - t
  w -= lr * error * x  (SGD step)
  e(x) = sigmoid(w . x)  (updated propensity)

Complexity: O(d) per update
Batch equivalent: O(n*d) for full logistic regression
```

2.2 Incremental DR Estimation

```
Maintain running sums:
  S_t = sum of T*Y/e(X) over all treated observations
  S_c = sum of (1-T)*Y/(1-e(X)) over all control observations
  n_t, n_c = counts
  sum_y_t, sum_y_c = sum of outcomes
  sum_y2_t, sum_y2_c = sum of squared outcomes

On new (x, t, y, e):
  S_t += t * y / e  (if treated)
  S_c += (1-t) * y / (1-e)  (if control)
  ACE = S_t/n_t - S_c/n_c
```

```
Complexity: O(1) per update
```

2.3 EWMA Drift Detection

```
EWMA (exponentially weighted moving average):
  ewma_ace = alpha * new_ace + (1-alpha) * old_ewma
  ewma_var = alpha * new_var + (1-alpha) * old_var

Drift detected when:
  |new_ewma - old_ewma| > 3 * sqrt(ewma_var)

alpha = 0.1 (default, 10-observation effective window)
```

3. Convergence Properties

Online estimates converge to the true causal effect as more data arrives. The self-test demonstrates this convergence:

```
After 50 observations:
  f0: ACE=1.000 SE=0.000 (true cause)
  f1: ACE=0.147 SE=0.132 (borderline, included)

After 100 observations:
  f0: ACE=1.000 SE=0.000 (stable)
  f1: ACE=0.080 SE=0.096 (drops below threshold)

After 200 observations:
  f0: ACE=1.000 SE=0.000 (converged)
  f1: ACE=0.062 SE=0.066 (tighter, excluded)
```

Key observations: (1) f0 is consistently identified as causal from the start — its ACE is 1.0 because it perfectly determines the outcome. (2) f1 starts with a borderline ACE (0.147) but converges to 0.062 as more data arrives, eventually dropping below the 0.1 threshold. (3) Standard error decreases monotonically (0.132 → 0.096 → 0.066), showing increasing confidence. This is the online refiner's key property: estimates get better with more data, without re-running the full batch.

4. Conclusion

v6.15 adds online causal refinement to the substrate. The OnlineCausalRefiner updates causal estimates incrementally in O(d) per observation, using streaming logistic regression, incremental DR sums, and EWMA drift detection. The substrate now maintains an up-to-date causal model in production — no batch re-estimation needed.

The complete causal reasoning hierarchy now has 9 levels:

```
Level 0: States               (v6.5)
Level 1: Correlational Laws   (v6.5)
Level 2: Refined Laws         (v6.9)
Level 3: Transferred Laws     (v6.10)
Level 4: Causal Laws          (v6.11)
Level 5: Structural Causal    (v6.12)
Level 6: Counterfactual       (v6.13)
Level 7: Observational Causal (v6.14)
Level 8: Online Causal        (v6.15)
Level 9: Active Learning      (future)
```

Each level adds a capability. The online causal layer (v6.15) is the production-ready layer: it enables real-time causal model adaptation with O(d) per-observation cost, drift detection, and convergence guarantees. Combined with the

prior layers, the substrate now has a complete causal reasoning pipeline: discover (observational + interventional), refine (online), reason (counterfactual), and transfer (meta-learning).

What v6.15 Does NOT Claim

v6.15 does not claim online estimates are as accurate as batch estimates with the same data. SGD-based logistic regression may not converge to the exact MLE, and incremental DR uses simplified running sums. The advantage is speed, not accuracy — online estimates are good enough for real-time decisions, but batch estimation should be run periodically for calibration. v6.15 also does not handle concept drift (where the TRUE causal structure changes) — it detects estimate drift but cannot distinguish 'more data' from 'structure changed'. This is a candidate for v6.16.

Appendix A. File Inventory

Artifact	Location	Notes
Online causal refiner	rce/rce2_online_causal.py	Streaming SGD + incremental DR + EWMA drift
v6.15 tests	tests/test_rce2_online_causal.py	18 tests, all passing
This report	rce/RCE2_V615_REPORT.pdf	PDF

Appendix B. How to Reproduce

```
# Run all 215 tests
cd /home/z/my-project/download
python -m pytest tests/test_rce2.py tests/test_rce2_adversarial.py \
  tests/test_rce2_substrate.py tests/test_rce2_density_gate.py \
  tests/test_rce2_adaptive_gate.py tests/test_rce2_reuse_classifier.py \
  tests/test_rce2_law_refiner.py tests/test_rce2_meta_learner.py \
  tests/test_rce2_causal_discovery.py tests/test_rce2_scm_discovery.py \
  tests/test_rce2_counterfactual.py tests/test_rce2_observational_causal.py \
  tests/test_rce2_online_causal.py

# Quick self-test
python -c "from rce.rce2_online_causal import self_test; self_test()"
```